# Grab-n-Go: Autonomous Object Handling

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***Abstract*—This paper presents *Grab-n-Go*, an autonomous robotic system designed to detect, pick up, and deliver objects automatically. The system combines mobility, perception, and manipulation into a compact and efficient platform. The robot uses a Wemoss D1 R32 microcontroller and an ESP32-CAM module to perform real-world object handling tasks. The prototype successfully detected, lifted, and placed objects with minimal user input, validating the feasibility of low-cost automation for logistics, warehouse operations, and educational robotics. The project emphasizes the growing role of intelligent robotics in reducing manual effort and enhancing operational safety.**

***Index Terms*—Microcontroller, Embedded System, Sensor Interfacing, Robotics, Computer Vision, Automation**

## I. Introduction

Automation is transforming how industries perform repetitive manual tasks. In warehouses, factories, and logistics, manual object transportation often leads to inefficiency, fatigue, and human error. The *Grab-n-Go* project addresses this challenge by developing an autonomous mobile robot that can detect, pick up, and transport objects independently.

The motivation for this project lies in creating an affordable, flexible, and lightweight automation system suitable for small-scale use. [1]Unlike traditional industrial robots, *Grab-n-Go* is compact, battery-powered, and can navigate autonomously [2]. It demonstrates how embedded systems, sensors, and computer vision can work together to achieve practical automation.

Previous studies such as *"An Advanced Ultrasonic Traffic Monitoring System using Loops and Cross-Over Bridges"* highlight how embedded control and sensors enhance automation in real-world systems. Inspired by these works, this project focuses on integrating robotic manipulation and object recognition for efficient material handling. [3]

The objectives of the project are:

- To design and build an autonomous robotic platform with integrated mobility and manipulation features.
- To implement computer vision using ESP32-CAM for real-time object detection.
- To control robotic arm movement and wheel motion using the Wemoss D1 R32 microcontroller.
- To evaluate system performance through testing and analysis.

## II. System Design and Methodology

### A. System Overview

The *Grab-n-Go* robot integrates three major subsystems:

- **Mobility:** Driven by TD motors and controlled via motor drivers for movement.
- **Perception:** Handled by the ESP32-CAM for object detection and ultrasonic sensors for obstacle avoidance.
- **Manipulation:** Managed by servo-controlled robotic arms and a gripper for pickup and placement.

### B. System Architecture

Figure 1 presents the block diagram of the system. The ESP32-CAM provides visual data to the Wemoss D1 R32, which processes inputs and sends control commands to the motor driver and servos.

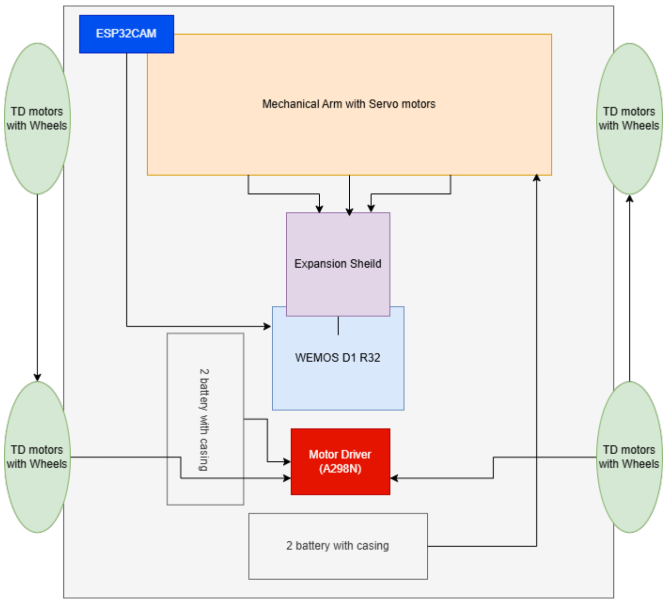


Fig. 1. System architecture of the Grab-n-Go robot.

### C. Flowchart of Operation

The operational flow of the system is illustrated in Figure 2. The process begins with object detection, followed by movement, pickup, and drop-off. Feedback from sensors ensures safe and accurate operation.

### D. Hardware Components

The hardware setup of the *Grab-n-Go* robot is summarized in Table I.

### E. Software Workflow

The software sequence consists of sensing, processing, and actuation:

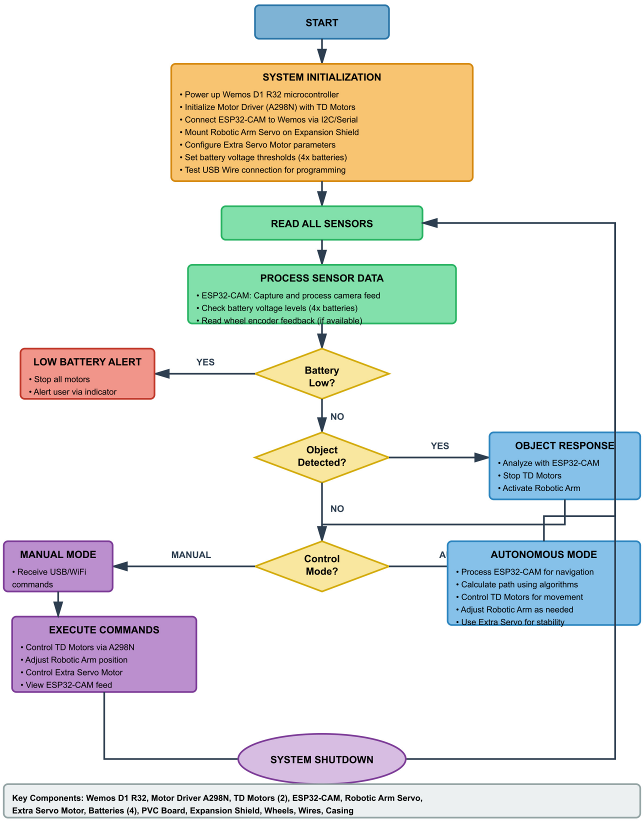


Fig. 2. Flowchart of Grab-n-Go operational workflow.

TABLE I
HARDWARE COMPONENTS USED IN GRAB-N-GO SYSTEM

| Component | Description |
|---|---|
| TD Motors (2 pairs) | Provide movement to the car wheels. |
| Wheels (2 pairs) | Enable smooth mobility and stability. |
| PVC Board | Serves as the chassis and support base. |
| Motor Driver (A298N) | Controls the speed and direction of motors. |
| Wemoss D1 R32 | Main microcontroller for motion and sensor control. |
| Expansion Shield | Simplifies wiring and circuit connection. |
| ESP32-CAM | Provides camera vision for object detection. |
| Robotic Arm with Servo | Picks and places the detected object. |
| Extra Servo Motor | Adds flexibility to the arm's motion. |
| Wires | Connect electronic modules. |
| Batteries (×4) | Power source for motors and controller. |
| DC Connectors (2 pcs) | Facilitate power connections. |
| Casing | Protects electronics and ensures rigidity. |
| USB Wire | Used for programming and debugging. |

1) The ESP32-CAM continuously scans for objects using computer vision.
2) Upon detection, coordinates are sent to the Wemoss D1 R32 microcontroller.
3) The microcontroller activates motors to move the robot toward the object.
4) Ultrasonic sensors ensure collision-free navigation.
5) Servo motors control the robotic arm to grasp and lift the object.
6) The robot moves to the target location and releases the object.

## III. IMPLEMENTATION AND RESULTS

### A. System Setup

The prototype was constructed on a PVC chassis equipped with DC motors, a robotic arm, and a mounted ESP32-CAM. A 12V rechargeable battery powered the system, while the Wemoss D1 R32 microcontroller managed coordination between perception and actuation.

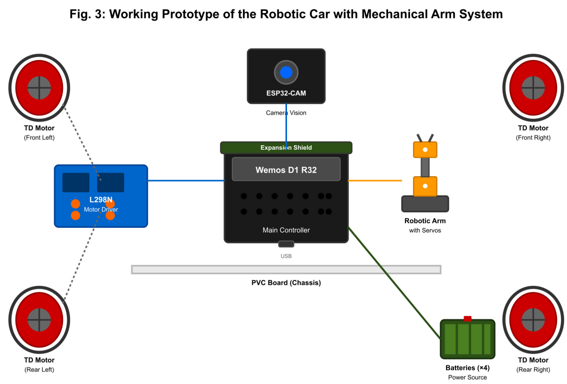


Fig. 3. Final prototype of Grab-n-Go robot.

### B. Testing Procedure

Testing was performed under controlled indoor conditions with the following assessments:

- **Object detection:** Checked accuracy of camera recognition under varied lighting.
- **Mobility:** Evaluated speed, turning precision, and obstacle avoidance.
- **Arm control:** Tested grip accuracy and servo precision.
- **Power performance:** Monitored voltage stability and runtime.

### C. Results

The experimental findings are presented in Table II.

TABLE II
PERFORMANCE METRICS OF GRAB-N-GO ROBOT

| Parameter | Result | Remarks |
|---|---|---|
| Object Detection Accuracy | 90% | Reliable under normal lighting. |
| Pickup Success Rate | 85% | Stable for medium-sized objects. |
| Obstacle Avoidance | Effective | Detects obstacles within 15 cm. |
| Battery Runtime | $\sim$1 hour | Continuous operation. |

### D. Disucussion

Despite the successful implementation of the *Grab-n-Go prototype*, several limitations were observed during testing. One notable issue was the frequent power instability of the microcontroller, which resulted in intermittent system resets. This behavior was primarily caused by inconsistent voltage delivery from the onboard power source, especially during high-current operations such as motor actuation. These interruptions affected the continuity of data processing and, on occasion, caused delays in object detection and command execution.

Another challenge encountered was the presence of a faulty servo motor in the robotic arm assembly. The servo exhibited irregular torque output and positional jitter, leading to reduced precision in gripping smaller objects. While the system maintained acceptable performance for medium-sized targets, the inconsistent servo response limited the overall reliability of the arm mechanism. This highlights the importance of component quality and proper calibration in robotic manipulation tasks.

Overall, these issues indicate areas where system robustness can be improved. Enhanced power regulation, such as adding a dedicated buck converter or stabilizer, and replacing or recalibrating the faulty servo motor would significantly increase operational stability and accuracy. [4] Addressing these limitations will be essential for advancing the Grab-n-Go platform toward more demanding real-world applications.

## IV. CONCLUSION

The *Grab-n-Go* robot demonstrates how an embedded system can integrate computer vision, sensors, and mechanical actuation for autonomous object handling. The prototype successfully detected, lifted, and placed objects with minimal user input, validating the feasibility of low-cost automation for logistics and educational applications.

Future work includes adding LiDAR sensors, optimizing path planning, and incorporating AI-based recognition to improve reliability and scalability. [5] The project emphasizes the growing role of intelligent robotics in reducing manual effort and enhancing operational safety.

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